ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C2 · MIXTURES

Choose a method

★☆☆
Grade 4

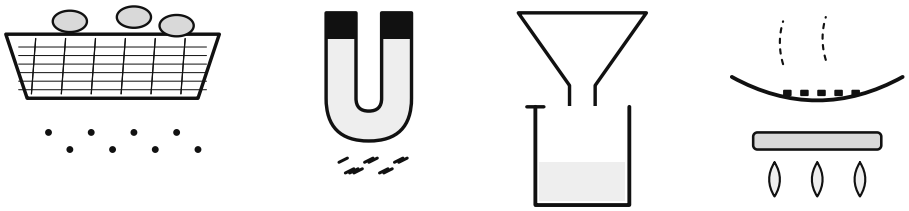
Learning goals

- Match mixtures to separation methods.
- Describe mixing.

1 Which method separates each mixture? [5]

Mixture	Method
stones in rice	picking out or sieving
iron pins in sand	using a magnet
sand in water	filtering (or pouring off)
salt in water	evaporating
husks and rice grains	winnowing

2 Name each method. [4]



A B C D

A	B	C	D
sieving	using a magnet	filtering	evaporating

3 What happens? Write dissolves, sinks, floats or fizzes. [4]

Mixture	What happens
salt in water	dissolves
a stone in water	sinks
oil in water	floats
vinegar and baking soda	fizzes

4 **Put the steps of recycling a metal can in order.** [3]

5	take out the new pot
1	collect old cans
4	let it cool and become solid
2	heat them until they melt
3	pour the liquid metal into a mould

ANSWER KEY

InquiryMind
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THINK IT THROUGH · C2 · MIXTURES

Separation puzzles

★★☆
Grade 4

Learning goals

- Plan how to separate a mixture.
- Explain using properties.

1 **Mina spilled a mixture of sand, salt, iron pins and dried peas. Plan how to get each one back. Explain each step.** [5]

1 Pick out or sieve the big peas. 2 Use a magnet for the iron pins. 3 Add water and stir: the salt dissolves. 4 Filter: the sand stays in the paper. 5 Evaporate the water to get the salt.

2 **Why can't you separate salt from water with a sieve or a filter?** [2]

The salt has dissolved into tiny pieces that pass through the holes with the water.

3 **Why must a mould be made of something with a much higher melting point than the metal poured into it?** [2]

Otherwise the hot liquid metal would melt the mould.

4 **Is mixing vinegar and baking soda a good way to make something you can separate again? Explain.** [1]

No: a new gas is made and escapes; you cannot get the vinegar and baking soda back.

ANSWER KEY

INVESTIGATE · C2 · MIXTURES

Which filter cleans muddy water best?

★★☆ Grade 4

Learning goals

- Compare materials fairly.
- Use a rating scale.
- Draw conclusions.

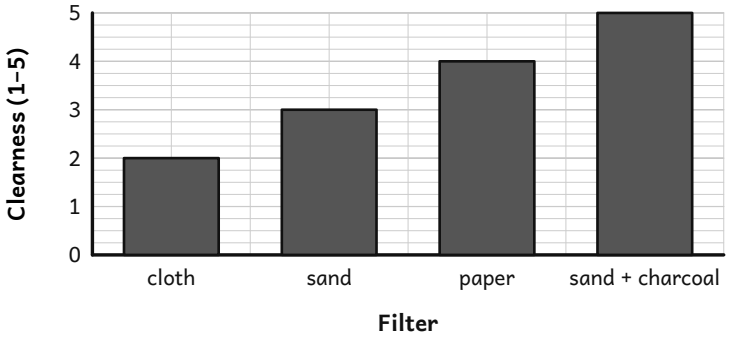
Class 4C poured 100 ml of the same muddy water through four different filters in funnels. They rated how clear the water was from **1** (still very muddy) to **5** (very clear).

1 **Complete the plan.** [3]

Change	the filter material
Measure	how clear the water is (rating 1–5)
Keep the same	the amount and type of muddy water, the funnel, the amount of filter material

2 **Draw a bar chart.** [3]

Filter	cotton cloth	sand	filter paper	sand + charcoal
Clearness (1–5)	2	3	4	5



3 **Which filter was best? Suggest why.** [2]

Sand and charcoal: the tiny gaps between sand grains trap mud and the charcoal traps even smaller bits.

4 **The water from the best filter looks clear. Is it safe to drink? Explain.** [2]

Not necessarily: germs and dissolved poisons like arsenic are too small to see and can pass through; it should be boiled or treated.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · C2 · MIXTURES

The SONO filter: clean water from a Bangladeshi invention

★★★
Grade 4

Learning goals

- Apply filtering to a real problem.
- Understand how a scientist designed a solution.
- Evaluate the solution.

Science · Health · Design · Economics

Millions of tube-wells in Bangladesh give water that is free of the germs found in ponds. But scientists discovered that the water from many wells contains **arsenic**, a poison that dissolves into the water from rocks deep underground. You cannot see, smell or taste it, and drinking it for years can cause serious illness.

Dr Abul Hussam, a chemist born in Kushtia, designed a simple filter called **SONO**. Water trickles through buckets filled with layers of **sand**, **brick chips**, **charcoal** and a special **iron mixture** that grabs the arsenic. It needs no electricity. In 2007 his invention won a major international prize for engineering, and many thousands of families have used it.

Based on information about arsenic in groundwater in Bangladesh and the SONO filter.

1 Match each filter layer to its job. [3]

Draw a line to match each pair, or write the letter.

sand and brick chips ●	→ C	● **A** grabs the dissolved arsenic
iron mixture ●	→ A	● **B** removes smells and tiny bits
charcoal ●	→ B	● **C** trap bits of mud and sand

2 Why can't ordinary cloth filter remove arsenic? [2]

The arsenic is dissolved, so it is in pieces far too small to be trapped by cloth.

3 Give two reasons why the SONO filter is a good design for villages. [2]

It needs no electricity; it is made from cheap local materials; it is simple to use and repair.

4 **What might each person think?** [3]

Person	View
a mother	I want safe water for my children without walking far.
a scientist	We must test each well and check the filters work.
a shopkeeper selling filters	Filters must be cheap enough for poor families to buy.

ANSWER KEY

InquiryMind
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REVIEW AND CHALLENGE · C2 · MIXTURES

Show what you know

★★★
Grade 4

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

M	M	O	J	S	I	I	F	V	S
D	E	V	P	A	C	S	I	L	I
I	O	L	V	L	M	P	L	A	E
S	O	L	T	T	N	D	T	H	V
S	M	I	X	T	U	R	E	M	E
O	S	T	N	W	W	I	R	A	L
L	A	O	K	O	H	L	V	G	E
V	N	A	S	V	R	I	K	N	W
E	D	V	L	W	P	C	L	E	D
M	O	U	L	D	E	Y	N	T	N

dissolve filter magnet melt mixture mould salt sand sieve

2 Which method gets salt back from salty water? [1]

- ○ sieving
- ○ filtering
- ● **evaporating**
- ○ using a magnet

3 A farmer tosses rice on a *kula* in the wind. This is called... [1]

- ● **winnowing**
- ○ filtering
- ○ melting
- ○ dissolving

4 Draw and label equipment to filter muddy water. [3]

Filtering

Funnel with filter paper, beaker underneath; mud left in the paper, clear water in the beaker.

5 **Challenge. Explain why recycling a metal pot by melting it is a reversible change, but burning paper is not.** [2]

Melting is a change of state: the liquid metal can become solid again. Burning makes new substances (ash, smoke, gases) so the paper cannot be got back.